PARASITIC LID DISEASE

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DEMODEX INFESTATION/DEMODICOSIS aka MITES

- Eyelid (also facial hair and eyebrow) mite infestation
- Prevalence
  - 0-25 years: 29%
  - 26-50 years: 53%
  - 51-90 years: 67% → ubiquitous
- Contributing factors
  - Mother-infant transmission (nursing)
  - Compromised immune system
  - Vector theory of spreading staph infection?

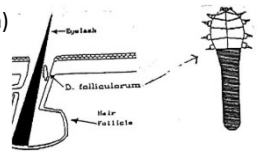


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Demodex infestation/Demodicosis

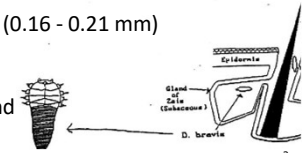
Demodex folliculorum

- Cigar shaped (0.28 - 0.44 mm)
- Colonial (3+ per follicle)
- Upper section of hair follicle
- Collarettes/cuffs
  - Best clinical diagnostic sign



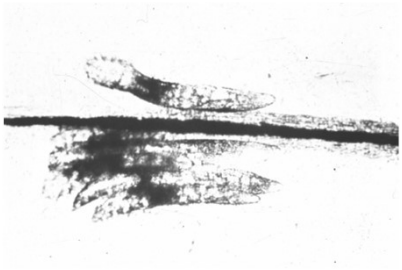
Demodex brevis

- Shorter, stubbier species (0.16 - 0.21 mm)
- Solitary
- Sebaceous glands
- Damage/plugging of gland
  - Dry eye
  - Chalazion formation



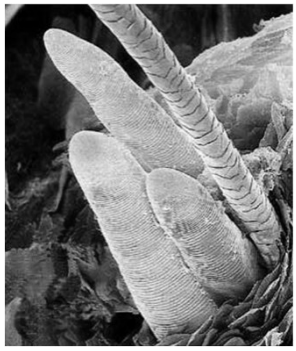
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Demodex Folliculorum



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Demodex Folliculorum

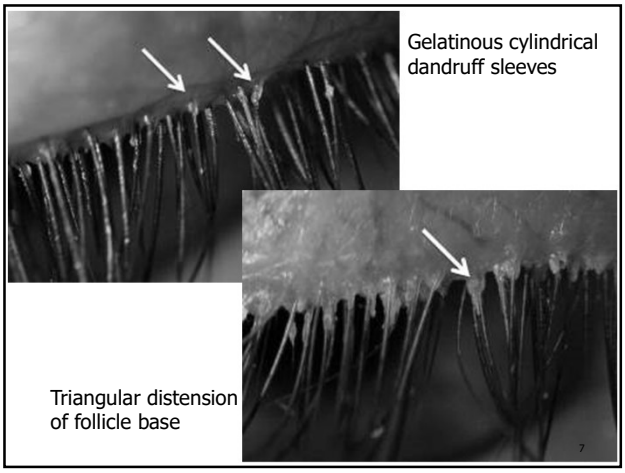


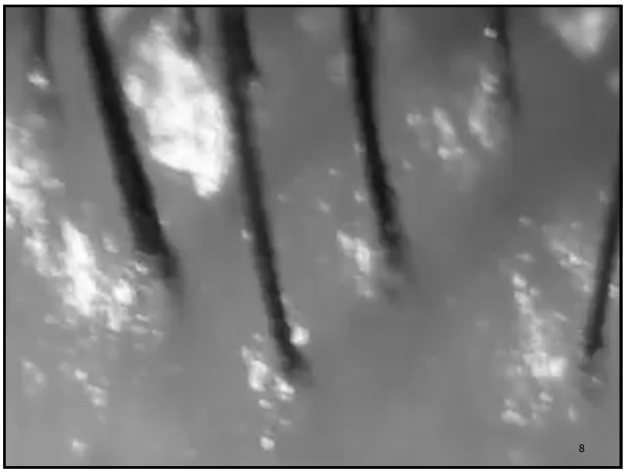
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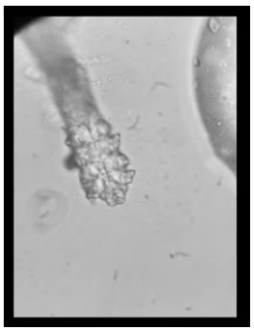
Demodex Lifecycle and Behavior

- Life cycle between 14 to 18 days
  - Egg → larva → adult
- Female mites lay up to 10-12 eggs at a time
  - Leads to overpopulation if unmanaged
- *Demodex* avoids light; buries head-first in follicles or glands, with only tail visible at lash follicle opening
  - Lays eggs, ingests gland lining
  - Results in follicular base distension and trichiasis
- Debris generated collects at lash roots
  - Appears as gelatinous cylindrical 'sleeve' or cylindrical dandruff (CD) at eyelash base (pathognomonic for mite infestation)

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https://contactlensupdate.com/2015/12/08/demodex-folliculorum/

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Demodex infestation/Demodicosis

Subjective

- Itching – especially in morning
- Burning
- Crusting
- Swelling of lid margins
- Lash loss

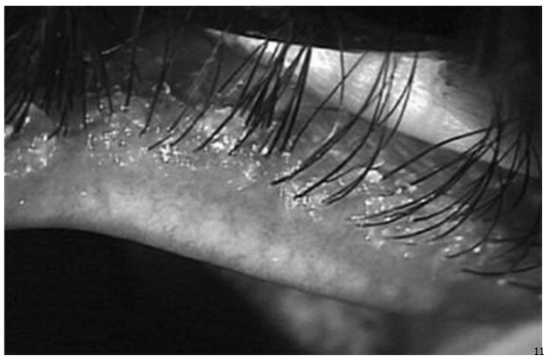


Objective

- Lid margin erythema
- Conj. injection
- Blepharitis
- Cuffing around lashes
- Plugging of meibomian glands
- Lash loss = _____

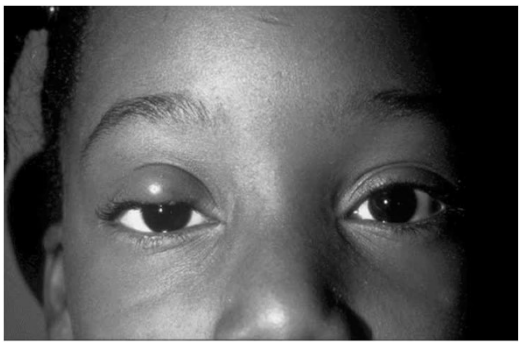
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Demodex Folliculorum



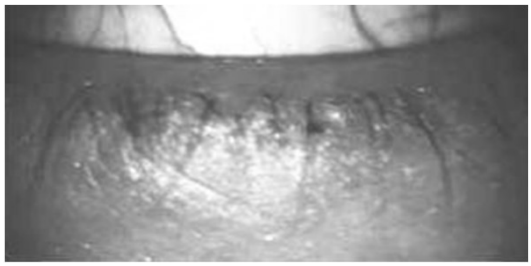
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Chalazion



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Madarosis



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Demodex infestation/Demodicosis

Assessment/DDX

- Case HX
- Patient's age
- Clinical signs



Plan/TX

- In-office procedure – aggressive lash scrubs
- Lid hygiene – Tea tree oil products may work
- Ung hs x 10-14 days (bland & viscous)
- Anti-microbial TX – for any super-infections

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Management



Tea tree oil derived from leaves of *Melaleuca alternifolia*, an Australian tree



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Cliradex (Bio-Tissue)

- Natural, preservative-free lid, lash, facial cleanser
- Has specific tea tree oil component (4-Terpineol)
  - More potent miticidal than tea tree oil as a whole
  - May reduce adverse/antagonistic effects of other components of tea tree oil
  - Scientifically proven to eliminate Demodex



- Safe to use daily; eliminates signs and symptoms
- Towelette qAM & qHS x 2 wks, then HS till no signs
- Clean eyelids/margins/eyelashes/periocular area
- Keep eyes closed for 1 min while areas air dry

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Ocusoft Oust Demodex Swabstix™

"Utilizing a unique blend of all natural oils including tea tree oil, sea buckthorn and coconut oil, Oust™ Demodex® Swabstix™ aid in the removal of adult Demodex mites and their offspring when used on the skin and skin structures surrounding the eye." (Ocusoft)



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Xdemvy



- Tarsus Pharmaceuticals
- lotilaner 0.25% ophth. sol.
- FDA approved July 2023
  - Saturn-1 and Saturn 2 trials
  - About 50% patients had reduction to 2 or less collarettes, and about 60% patients had *Demodex* mites completely eradicated by 6 weeks
- 1 drop both eyes twice daily (q12hr) for 6 weeks
- Sting/burn 10%; punctate keratitis/stye <2%

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Demodex treatment considerations

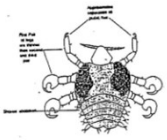
- Mites lifespan: 2-3 weeks
- Female will lay about 20 eggs in single lash follicle in that time
- Patients need several weeks (6-8 weeks in some cases) of treatment and maintenance
- Immunocompromised patients may retain more mites and for longer time periods
  - Mites may produce immunologic suppressants

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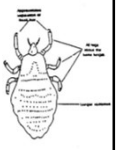
PHTHIRIASIS PALPEBRARUM*/ PEDICULOSIS PALPEBRARUM aka LICE

Eyelid infestation with lice

1. *Phthirus pubis* (crab louse)
2. *Pediculus corporis* (body louse)
3. *Pediculus capitus* (head louse)



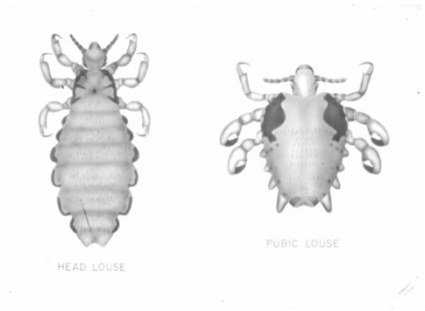
- Vector of serious diseases (typhus)
- Free-moving (*Pediculus*) vs. localized (*Phthirus*)
- Nits (eggs)
- Phthiriasis may be considered a STD



*also called phthiriasis ciliaris or ciliary phthiriasis

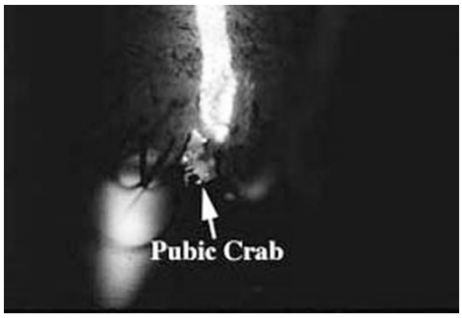
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Head & Pubic Lice



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Phthiriasis pubis



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Phthiriasis pubis



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Phthiriasis pubis

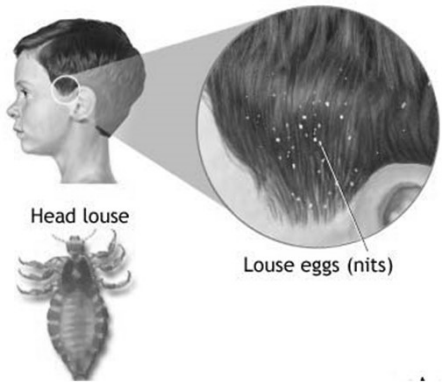


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Pediculosis corporis

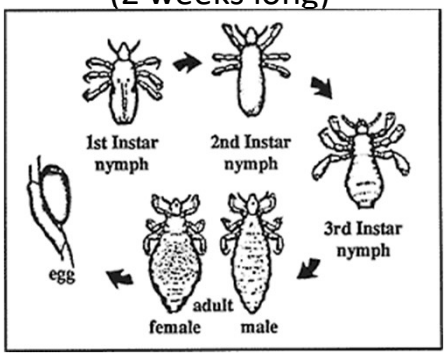


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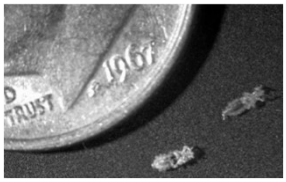
Pediculosis capitus life cycle (2 weeks long)



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Pediculosis capitus

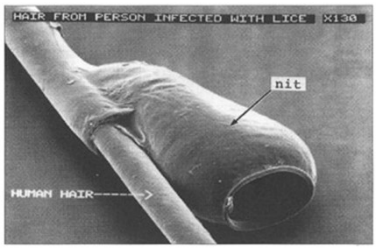




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Pediculosis capitus





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PHTHIRIASIS PALPEBRARUM/ PEDICULOSIS PALPEBRARUM aka LICE

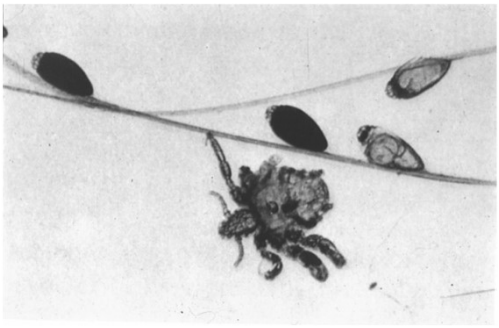
Subjective
- Itching
- Red eye
- Insomnia, irritability
- Occasional preauricular lymphadenopathy

Objective
- Nit cases
- Crusty lid margins
- Madarosis
- Conj injection
- Lid edema, erythema
- Adults



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Phthiriasis



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Phthiriasis pubis



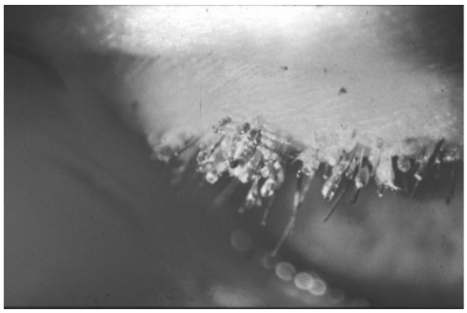
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Phthiriasis



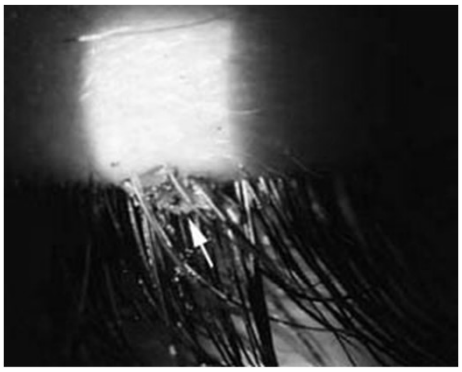
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Phthiriasis pubis



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Phthiriasis pubis



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PHTHIRIASIS PALPEBRARUM/PEDICULOSIS PALPEBRARUM aka LICE

Plan/TX

- In-office procedure (mechanical removal)
- Ung tid x 10-14 days for OCULAR tx
  - erythromycin, tobramycin, bacitracin, other ABs
  - Prasympathomimetics: physostigmine 0.25% (Eserine) ung, 2 applications 1 week apart, or pilocarpine ointment
  - Vaseline, petrolatum ointment
  - Pediculocidal agents for NON-OCULAR areas
  - Cryotherapy and argon laser have been attempted

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PHTHIRIASIS PALPEBRARUM/ PEDICULOSIS PALPEBRARUM aka LICE

Topical anti-parastic agent options

BRAND	ACTIVE INGREDIENT	KILLS EGGS	RESISTANCE	PRICE
Nix, Rid, Pronto	Permethrin/ pyrethrin neurotoxin	No	Possibly	$10-$20
LiceMD	Dimethicone	No	No	$13-$15
Ovide RX	Malathion, neurotoxin	Yes	Not included	$125-$153
Ulesfia RX	Benzyl alcohol	No	No	$30

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Management



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PHTHIRIASIS PALPEBRARUM/PEDICULOSIS PALPEBRARUM aka LICE

Plan/TX – less conventional

- Alcohol-coated cotton swab lid margin epithelial debridement
  - care should be taken in this procedure
- Oral ivermectin (200-400 microgram/kg as single dose; second dose option 7-10 days later)
  - Off-label (i.e. not FDA-approved), but effective
  - Head lice treatment; avoid if pregnant/age 0-5 years

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PHTHIRIASIS PALPEBRARUM/ PEDICULOSIS PALPEBRARUM aka LICE

Other TX considerations:

- Referral
- Family members, sexual contacts
- Home environment
- Pt. education
- Abuse issues



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THANK YOU!



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